

Crowding the face-space: The attractor field hypothesis and within-person variability



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Background

Two sides of the face recognition problem:

- Telling people apart (differentiation)
- Telling people together (amalgamation)

Discrimination

The face space theory (Valentine, 1991)

- Faces are encoded in a multidimensional space according to inter-item similarity
- Typical faces are clustered around the central tendency, distinctive faces are encoded in isolated regions

Amalgamation

The attractor field model of the face space (Tanaka et al., 1998)

- Faces are surrounded by regions that allow for multiple inputs to activate a face's representation
- Typical faces have smaller attractor fields; distinctive faces have larger attractor fields
- The size of an attractor field can predict tolerance to within-person variability in the ORE (Laurence et al., 2016)



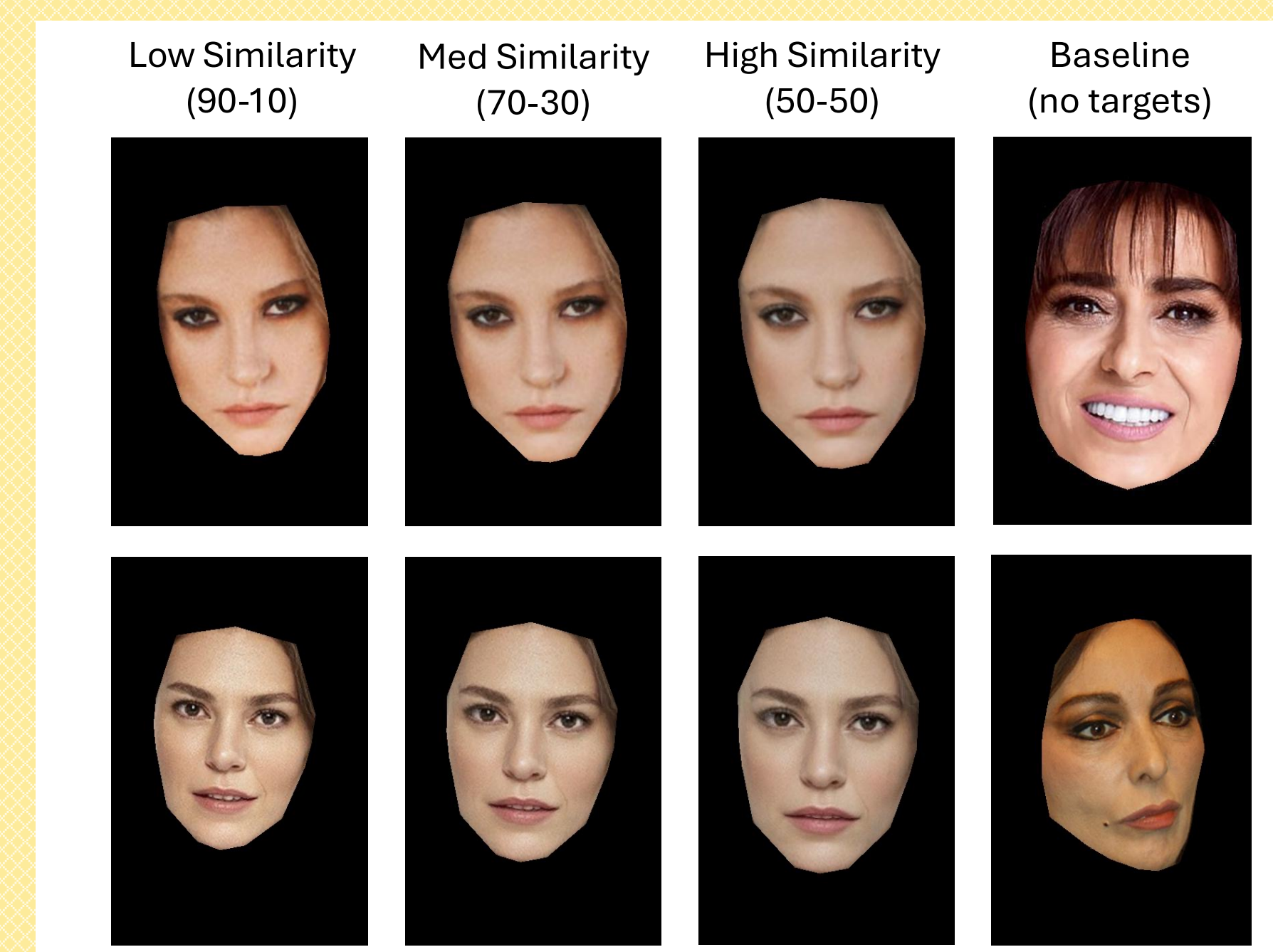
How does between-people variability influence learning a new face through exposure to within-person variability?

Hypotheses

- Tolerance to within-person variability (i.e., amalgamation) will be worse for faces that are encoded in higher-density regions
- Faces in higher-density regions could be more difficult to differentiate (the attractor field model) OR
- Faces in higher-density regions could be easier to differentiate (attention to distinguishing features)

Methods

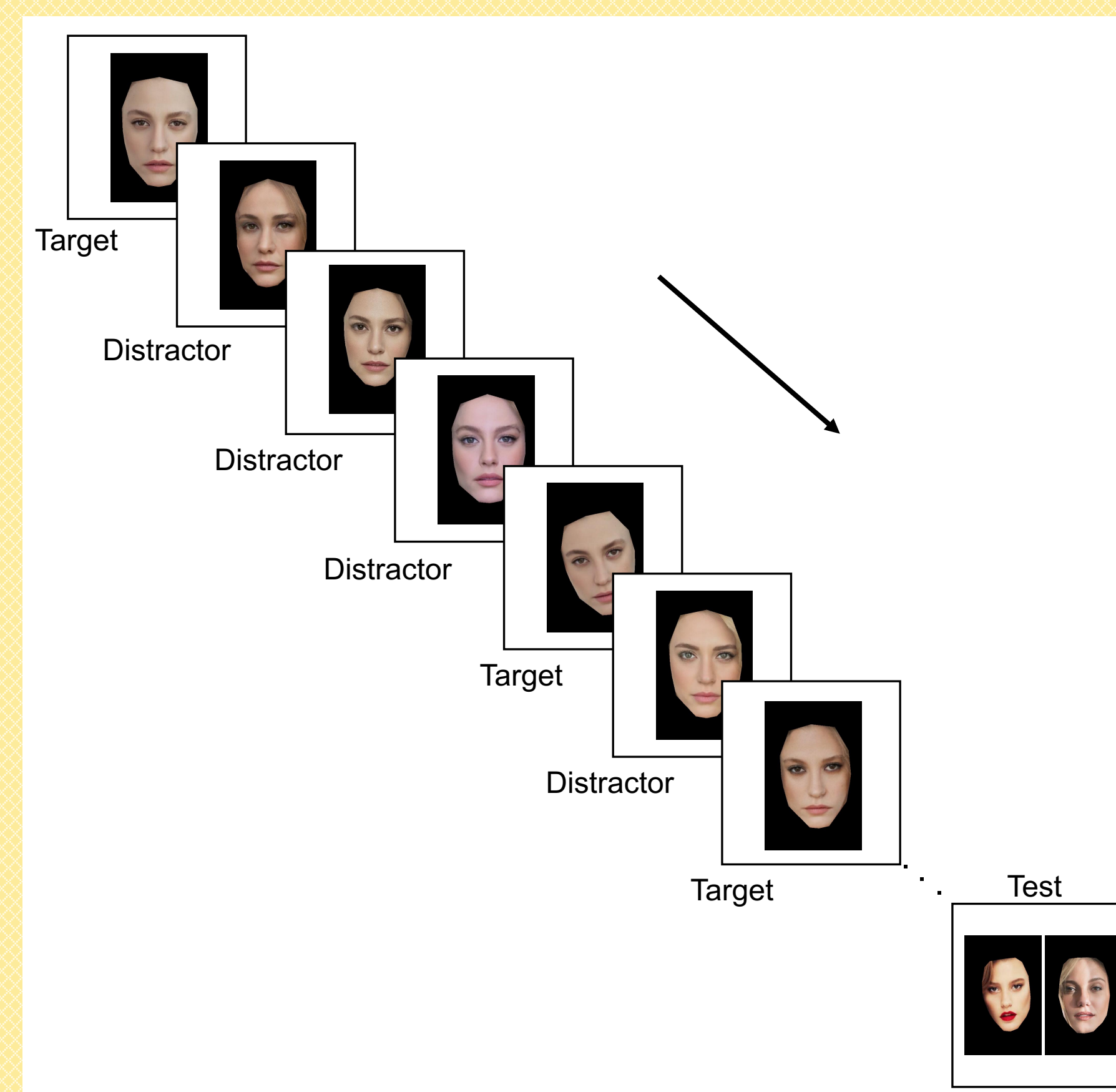
4 between-subjects conditions:



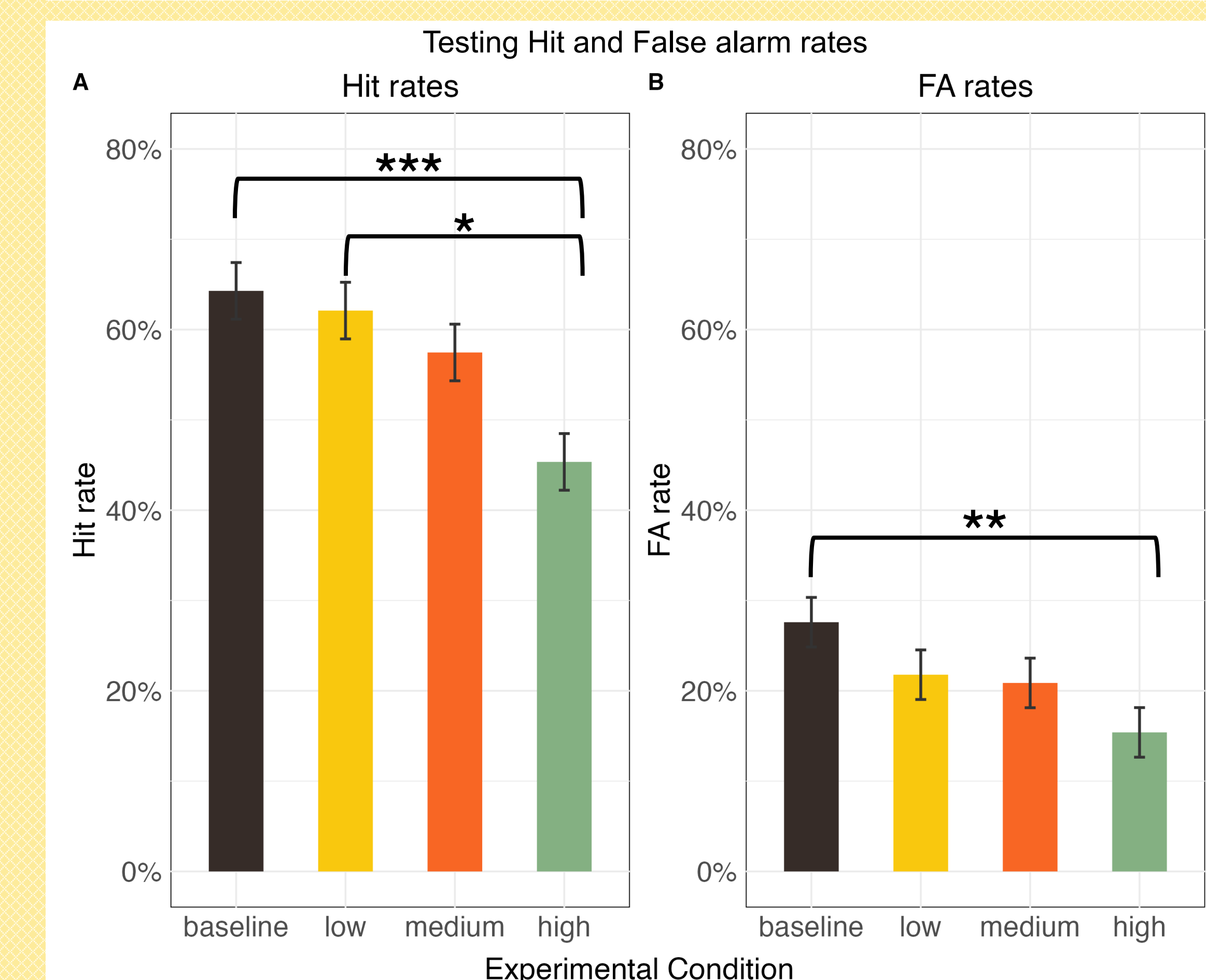
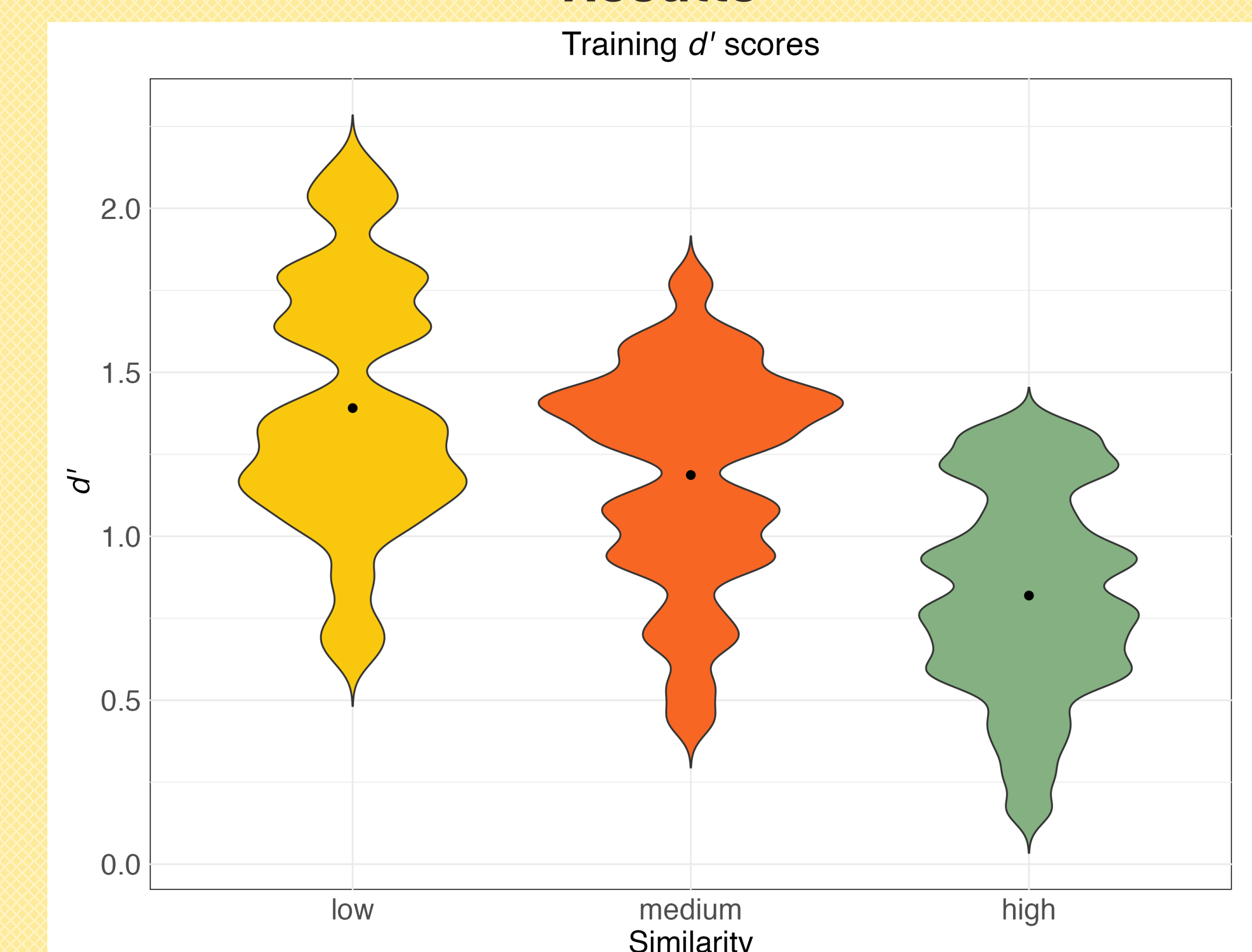
Training: Target classification task

- Baseline: gender classification task

Testing: Matching task



Results



N=186

Discussion

- Recognition sensitivity is worse for faces that are encoded in denser regions of the face space (i.e., high similarity).
- In line with the attractor field model, there is less tolerance to within-person variability (i.e., lower hit rates) for faces in high-density regions compared to lower-density regions.
- Contrary to the attractor field model, differentiation performance was better for faces that were in the highest density regions compared to baseline. This could be due to increased attention to distinguishing features of targets during training.
- Interestingly, despite prior exposure to within-person variability, amalgamation for faces that were in the highest density regions was worse than not being exposed to targets at all.

